

- 9 (a) By reference to the photoelectric effect, state what is meant by the *work function energy* of a surface.

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.....
..... [1]

- (b) Fig. 9.1 shows two parallel metal plates A and B enclosed in an evacuated glass tube. Plate A is at an electric potential of 3.7 V.

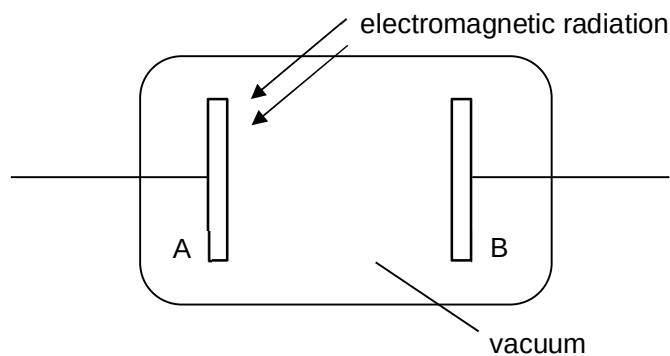


Fig. 9.1

- (i) Plate A is illuminated with electromagnetic radiation.

The variation with electric potential of plate B of the photoelectric current is shown in Fig. 9.2.

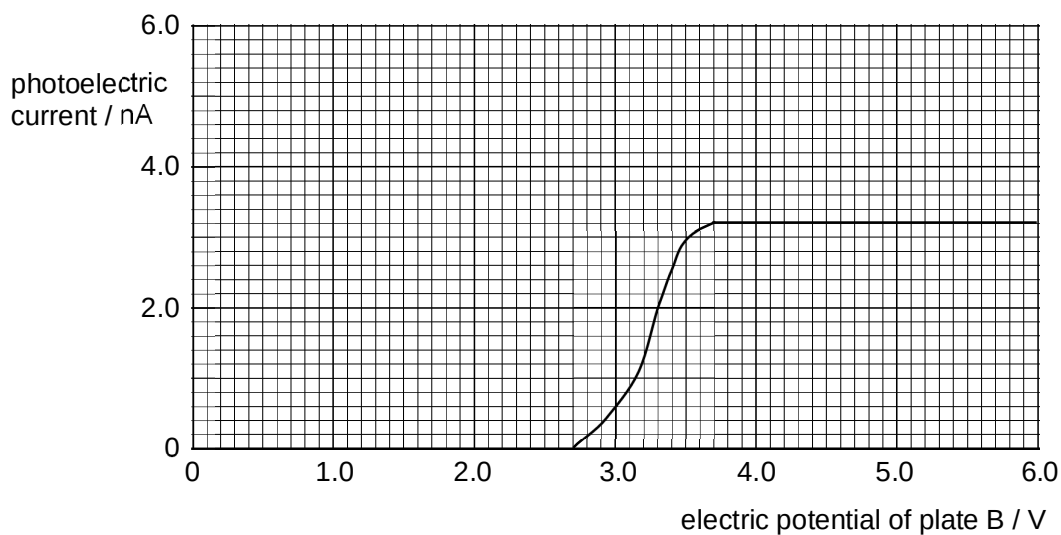


Fig. 9.2

1. With reference to photoelectrons, explain the significance of the sloping section of the graph in Fig. 9.2.

.....

 [2]

2. Calculate the maximum speed of the photoelectrons emitted from plate A.

maximum speed = m s^{-1} [3]

3. Calculate the number of electrons collected per second by plate B.

number of electrons per second = s^{-1} [2]

4. Calculate the number of photons incident per second on plate A, if one electron is emitted for every 1.0×10^5 incident photons.

number of photons per second = s^{-1} [1]

5. The frequency of the electromagnetic radiation is kept constant as its intensity is halved. On Fig. 9.2 sketch a graph to show the variation with electric potential of plate B of the photoelectric current for this decreased intensity. [2]

- (ii) Electromagnetic radiation of wavelength λ illuminates plate A. Electrons are emitted from plate A with maximum speed $v_{\max}$.

The variation with $\frac{1}{\lambda}$ of $v_{\max}^2$ is shown in Fig. 9.3.

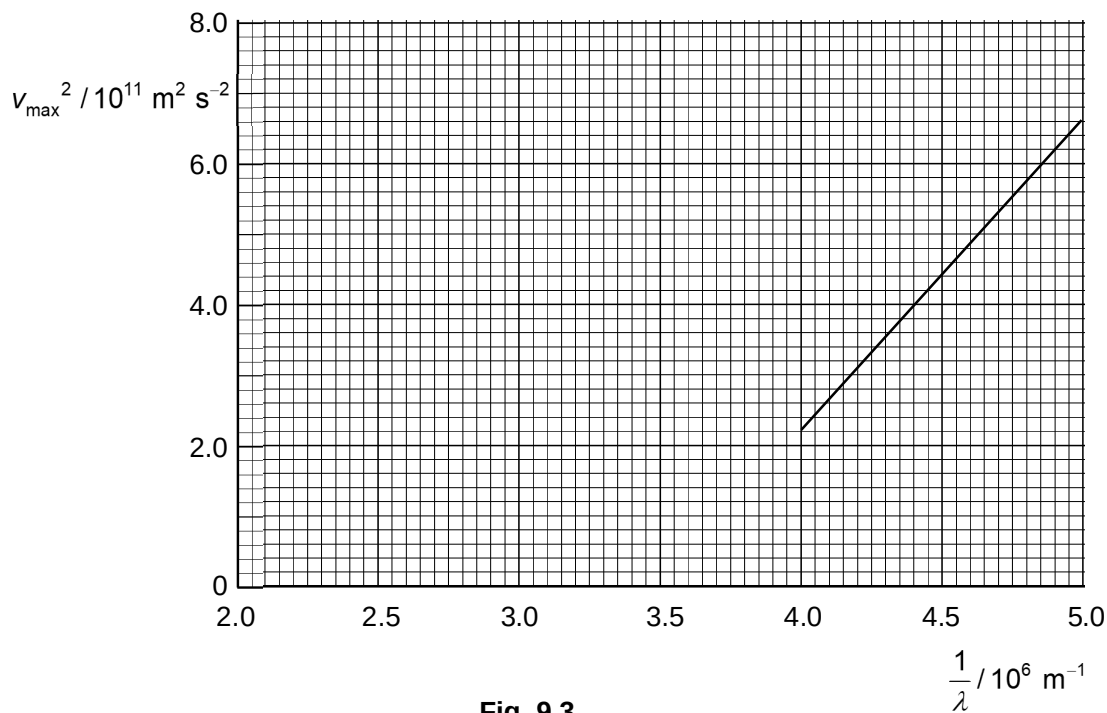


Fig. 9.3

1. Use Fig. 9.3 to determine the Planck constant.

Planck constant = J s [3]

2. Use Fig. 9.3 and your answer in 1. to determine the work function energy of plate A.

work function energy = eV [2]

3. On Fig. 9.3, draw a line to show the variation with $\frac{1}{\lambda}$ of $v_{\max}^2$ for a metal with a lower work function energy. [1]

(c) Electromagnetic radiation of wavelength 200 nm is emitted from a source rated at 0.76 mW. The photons are incident normally on a metal surface and are all absorbed by the surface.

- (i) Show that the number of photons incident per second on the metal surface is $7.6 \times 10^{14} \text{ s}^{-1}$.

[1]

- (ii) Calculate the magnitude of the force exerted on the metal surface by the incident photons.

force = N [2]

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